

SYNTRIXA

INTERNAL AGENT OS · v1

Architecture Decision Master
Summary

Complete System Design · Agent Specifications · Workflow Models
Memory Architecture · Execution Contracts · Deployment Strategy

CONFIDENTIAL — INTERNAL DOCUMENTATION — Syntrixa Org | AI Operations Team

System Version	v1 (Near Production)
Document Type	Architecture Master Summary
Agent Count	6 Specialist + 1 Supervisor
Audience	P1 · P2 · P3 · P4 · P5

This document is the definitive architectural reference for the Syntrixa Internal Agent OS v1. It captures all finalized decisions spanning agent design, workflow models, memory architecture, execution contracts, and deployment strategy. The system is designed as a near-production AI operational organization — not a prototype — where intelligent agents behave as specialized employees under a governed orchestration framework.

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01 · SYSTEM OVERVIEW & CORE PHILOSOPHY

What Syntrixa Agent OS is and the principles it's built on

System Goal

The Syntrixa Internal Agent OS is an AI-powered operational organization that uses intelligent agents, human collaborators, and automated workflows to run core business functions — including lead intake, qualification, research, marketing, communication, and system monitoring. The platform also serves as a live learning environment for building and operating production-grade orchestration infrastructure, from Docker and monitoring to auth and observability. Long term, this same framework will be adapted to build client-specific operational systems — not rigid templates, but custom-assembled agent architectures.

AI Employee Model	Agents reason, collaborate, communicate, escalate, learn, and maintain state. They are not simple prompt chains — they behave like specialized employees.
Domain Ownership Model	Each agent owns only its area of specialization. Work outside that domain is delegated to the correct specialist, ensuring precision and minimal hallucination.
Reasoning / Execution Separation	OpenClaw handles all reasoning and orchestration. n8n handles all execution. Agents never directly call APIs, send email, or mutate systems.
Framework Strategy	No universal template. A reusable framework is assembled into a custom architecture for each client deployment — the graph changes, the scaffolding reuses.
Near Production Commitment	The implementation targets near-production quality from the start — not a toy MVP or fake architecture, but a real operational system built incrementally.

Key Decisions at a Glance

Single Supervisor	One supervisor agent coordinates all routing, escalation, and lifecycle management.
Hybrid DB	MongoDB as primary v1 store; PostgreSQL added later for structured operational data.
Adaptive Context Transfer	Handoffs between agents use the minimum context the receiver actually needs.
No Autonomous Debates	Agent conflicts escalate to the Supervisor, which asks the appropriate human.
Channel Abstraction	Agents request communication abstractly — the system selects email, desktop alert, etc.
Role-Based Models	Different agents can run on different LLM models suited to their reasoning needs.

02 · FULL SYSTEM ARCHITECTURE

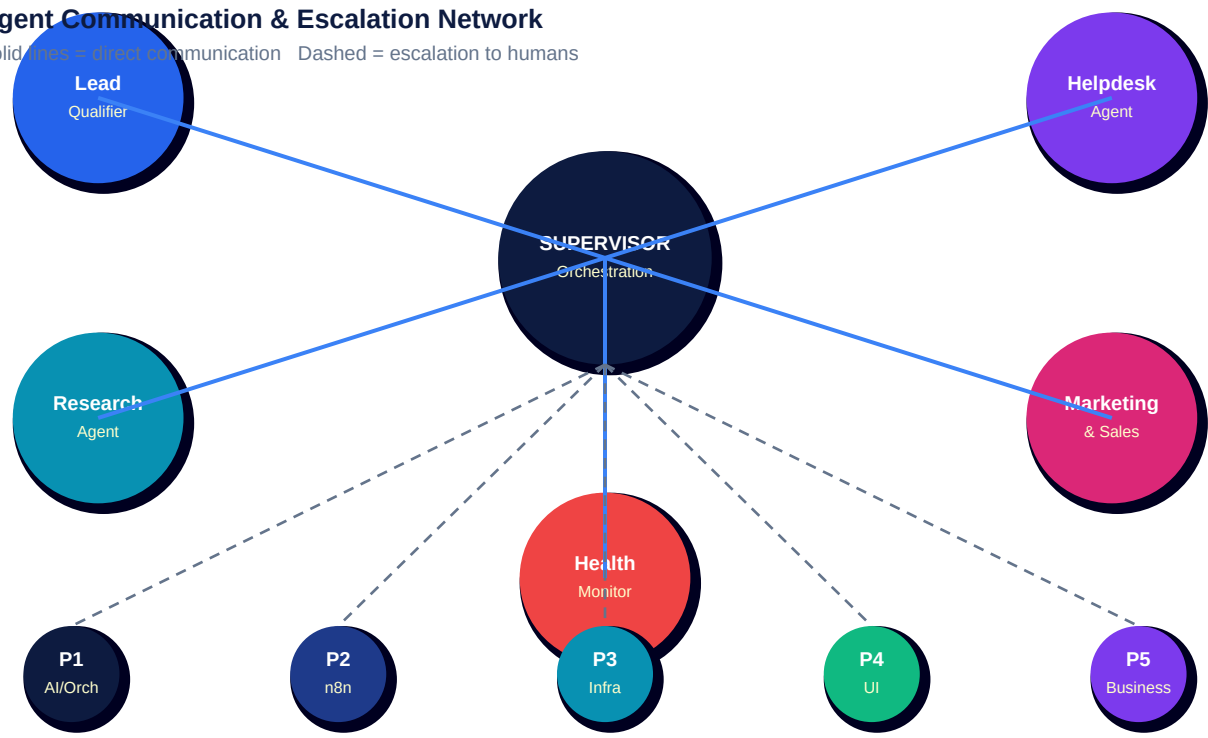
Layered view of every system component and how they interrelate

The architecture is organized into eight stacked functional layers. Each layer communicates bidirectionally with its neighbours. Humans sit at the top as integrated operational actors, not fallback systems. Execution is always the bottom concern, isolated in n8n so that reasoning agents never touch live infrastructure.



Agent Communication & Escalation Network

Solid lines = direct communication Dashed = escalation to humans



03 · AGENT SPECIFICATIONS

Complete profile for all 6 specialist agents + supervisor

The agent network is composed of one central Supervisor and five domain specialists. Each agent is a domain owner: it reasons deeply within its specialty and delegates everything else. All agents communicate through the Supervisor for coordination and route operational actions through n8n.

1 · Supervisor Agent

Primary Owner	P1
Recommended Model	Strong reasoning model
Execution Model	Mixed / event-driven
Purpose	Central coordinator. Routes tasks, tracks lifecycle, escalates to humans, and governs all inter-agent coordination. The Supervisor does not become a universal executor or knowledge sink.
Orchestration	Controls workflow progression, routing, coordination, and lifecycle assignment across the entire agent network.
State Awareness	Tracks active workflows, task states, lifecycle stages, escalations, and ownership — maintaining a minimal but accurate operational picture.
Escalation Management	When an agent faces uncertainty, disagreement, or requires human approval, the Supervisor coordinates the escalation to the right person.
Communication Mediation	Agents communicate intent; the Supervisor translates requests into the appropriate channel (email, desktop, future dashboard).
Memory Minimalism	Default mode is minimal awareness. The Supervisor reads agent memory only when coordination, debugging, or escalation requires it.

■ Boundaries: Not a universal executor, deep specialist, or omniscient reasoning sink.

2 · Lead Qualification Agent

Primary Owner	P5 (consumer), P1 (system)
Recommended Model	Conversational reasoning model
Execution Model	Event-driven (form submission, reply)
	Validates inbound leads — distinguishing genuine interest from spam — and builds sufficient context for the Research Agent. Does not consult, does not answer service questions.
Spam & Authenticity Detection	Uses form data, identity context, and prior history to determine if a submission represents a real, serious potential client.
Adaptive Qualification	Asks only the new questions needed to understand intent — not a rigid questionnaire. Uses prior context to avoid redundancy.

Context Building	Collects the information the Research Agent will need for deep intelligence gathering, nothing more.
Service Question Routing	Any question about Syntrixa services, capabilities, or offerings is immediately routed to the Helpdesk Agent — the Lead Qualifier never answers these.
Handoff to Research	Once sufficient understanding is achieved, transfers an adaptive context payload to the Research Agent.

■ Boundaries: No deep research, no marketing, no helpdesk consulting, no execution.

3 · Research Agent

Primary Owner	P5 (consumer), P1 (system)
Recommended Model	Strong analysis model
Execution Model	Event-driven (qualified handoff)

Purpose	Produces deep client intelligence. Consumes form data, the qualification conversation, and public sources to generate an executive brief and a full intelligence dossier delivered to P5.
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Business Intelligence	Analyzes company profile, business model, maturity, and industry context from public sources and the qualification payload.
Technical Inference	Infers the client's likely tech stack, workflows, and automation readiness based on available signals.
Opportunity & Bottleneck Mapping	Identifies likely operational pain points and maps automation opportunities specific to the client.
Lead Scoring	Generates lead score, estimated deal size, service fit assessment, and risk flags.
Dual Output Mode	Produces a concise executive brief (default) and a full expandable intelligence dossier, both delivered to P5 via the communication layer.

■ Boundaries: No lead qualification, no helpdesk, no marketing, no execution.

4 · Helpdesk Agent

Primary Owner	P5
Recommended Model	Fast conversational model
Execution Model	Event-driven (visitor question, routed question)

Purpose	Answers public knowledge questions about Syntrixa — services, capabilities, FAQs, and process guidance. Identity-aware: retrieves user context before every response to support conversation continuity.
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Identity-First Reasoning	Before responding, dynamically retrieves identity context from the DB: known lead? existing client? active workflow? prior request?
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Knowledge Communication	Answers questions about Syntrixa offerings, services, workflows, deployment models, and FAQs from the Central Knowledge Base.
Smart Guidance	Guides visitors toward the right service, suggests next steps, and navigates complex conversations without overstepping into consulting territory.
Conversation Continuity	Uses retrieved context to pick up existing threads seamlessly, making returning users feel recognized.
Routed Question Handling	Accepts questions forwarded by the Lead Qualifier and responds in context, returning the thread where appropriate.

■ Boundaries: Not a deep consultant. Not a researcher. Not a marketing agent.

5 · Marketing & Sales Agent

Primary Owner	P5 (operational), P1 (system)
Recommended Model	Growth/analysis model
Execution Model	Mixed — event-driven + scheduled
Purpose	Owns the growth domain. Analyzes campaign performance, generates growth intelligence, tracks sales signals, and proposes optimization actions — governed through the Supervisor.
Campaign Operations	Manages campaign creation, drafting, targeting, optimization proposals, and campaign lifecycle.
Performance Analytics	Tracks and analyzes CTR, CPC, CPA, conversions, and audience behaviour to generate actionable insights.
Growth Intelligence	Generates growth ideas, expansion suggestions, and audience insights informed by historical performance patterns.
Sales Intelligence	Tracks lead acquisition sources, service demand trends, channel quality, and high-value audience signals to inform business strategy.
Governed Semi-Autonomy	Can propose and initiate operational changes — but all actions are routed through the Supervisor and executed via n8n. No unrestricted autonomy.

■ Boundaries: No qualification, no research, no helpdesk.

6 · Health Monitoring Agent

Primary Owner	P3 (primary), P1/P2/P3 (coordination)
Recommended Model	Lighter diagnostic model
Execution Model	Mixed — failure-triggered + periodic checks
Purpose	Ensures operational reliability. Detects failures, attempts recovery, and escalates to the right human when needed. For critical security events, alerts Syntrixa support immediately.

Diagnostics	Detects workflow failures, API errors, stuck states, and degraded operations across the execution layer.
Recovery Attempts	Before escalating, attempts retries, corrective actions, and clarification gathering autonomously.
Supervisor Coordination	When recovery is insufficient, requests human intervention through the Supervisor, routing to P3 for infra issues or P2 for automation issues.
Critical Incident Path	For security events or severe failures, immediately alerts the Syntrixa support email without waiting for Supervisor mediation.
Operational Learning	Accumulates private memory of recovery patterns, failure signatures, and diagnostic heuristics over time.

■ Boundaries: No business reasoning, no marketing, no deep orchestration.

04 · WORKFLOW & INTERACTION MODELS

End-to-end flows for every primary operational path

Primary Contact Request Workflow

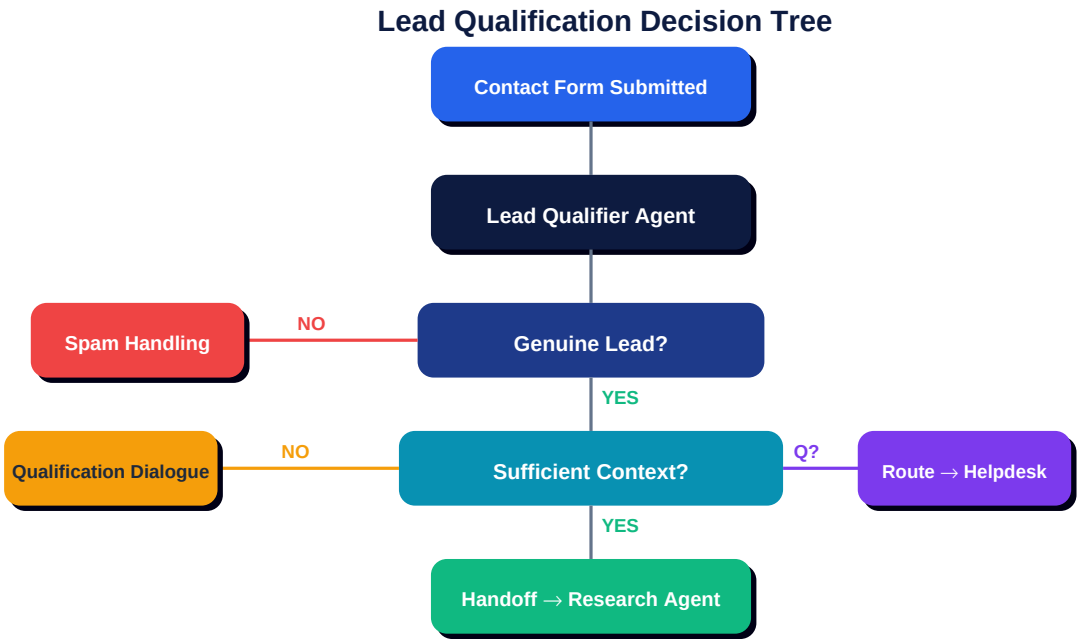
This is the core operational path — from a visitor submitting a contact form to P5 receiving a complete research dossier.

End-to-End Contact Workflow



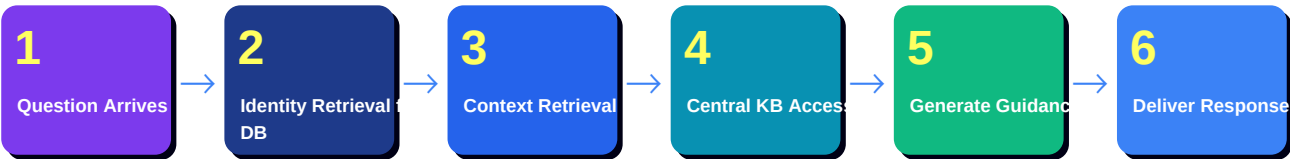
Lead Qualification Decision Tree

The Lead Qualifier applies a layered decision process. Any client question about services triggers an immediate route to the Helpdesk — the Qualifier never answers service questions itself.



Helpdesk Identity-First Workflow

Helpdesk Response Flow



Marketing & Sales Operational Workflow

Marketing → Sales Intelligence Flow



Health Monitoring Incident Workflow

Failure → Recovery → Escalation



■ Critical Incident Path — For security events or severe failures, the Health Agent bypasses the Supervisor and immediately alerts Syntrixa support mail.

Conflict Resolution Model

There are no autonomous agent debate loops. If a meaningful disagreement or uncertainty arises between agents, the Supervisor is immediately involved, and the Supervisor escalates to the appropriate human for a decision before the workflow continues.



05 · MEMORY, KNOWLEDGE & IDENTITY ARCHITECTURE

How agents store, retrieve, and share information

The system uses a Hybrid Adaptive Memory Model. Rather than a universal shared memory store, each agent maintains isolated private memory appropriate to its domain, with shared memory used only for operationally necessary cross-agent context. The Supervisor operates with minimal memory by default.

Hybrid Adaptive Memory Architecture



Memory Governance Rules

Working Memory	Task-scoped and disposable. Not automatically persisted. Holds active conversation, current workflow state, or live research session.
Long Memory	Stores compressed, distilled learnings — not raw transcripts. Agents retain useful patterns, heuristics, and organizational preferences.
Private Memory (per agent)	Fully isolated. Lead Agent holds spam/qualification patterns; Research holds intelligence techniques; Marketing holds campaign learnings; Health holds diagnostics history.
Shared Memory	Only for operationally necessary context: lead state, lifecycle stage, workflow state, or critical organizational decisions.
Supervisor Memory	Default minimal awareness. Reads agent memory only during coordination, escalation, or debugging — to avoid token overhead.
Learning Governance	Agents may ask 'Should I remember this?' if uncertain. Temporary preferences are discarded; organizational rules are stored; long-term patterns are retained appropriately.

Knowledge Architecture

Three knowledge tiers support agent reasoning: a central organizational KB, per-agent specialized knowledge, and a dynamic identity retrieval system.

Central Organizational KB	Agent-Specialized KB	Dynamic Identity Context
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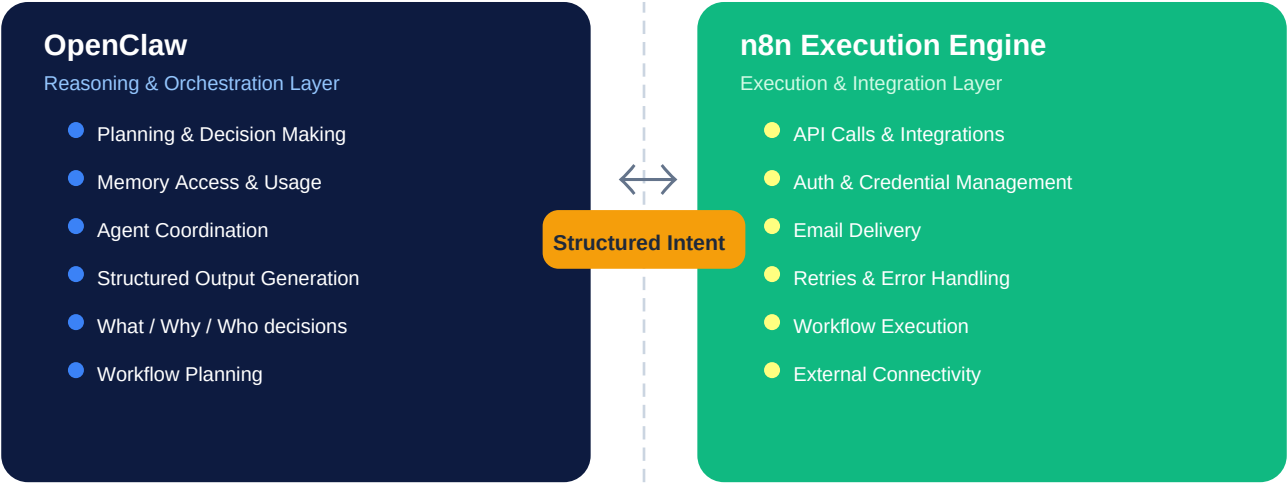
Services & offerings Workflows & processes FAQs Operational philosophy Used by: Helpdesk, Lead, Supervisor	Lead: qualification heuristics Research: intelligence methods Marketing: growth patterns Helpdesk: communication style Health: diagnostic knowledge	Pulled from DB before reasoning: Known user? Active lead? Prior request? Lifecycle stage? Assigned owner? Workflow state? Critical for: Helpdesk, Lead, Supervisor
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06 · OPENCLAW ↔ n8n EXECUTION CONTRACT

The core separation between reasoning and execution

The most important architectural rule in the system: OpenClaw reasons, n8n executes. No agent — regardless of domain or urgency — ever calls an API directly, authenticates with a service, sends an email, or mutates any external system. Every operational action is expressed as a structured intent, routed through the Supervisor, and executed by n8n.

OpenClaw ↔ n8n Separation Contract (Core Architectural Rule)



Execution Flow

Agent Reasoning → External Action



Structured Output Contract

Agents use natural language for reasoning, discussion, and clarification. For all actionable operations, structured JSON outputs are required to ensure reliable workflow triggering and auditing.

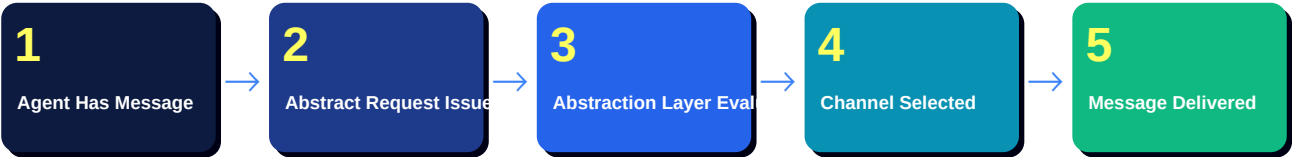
Workflow Request	{ "type": "workflow_request", "workflow": "deliver_research_summary", "priority": "normal" }
Handoff	{ "type": "handoff", "source": "Lead", "target": "Research" }
Human Escalation	{ "type": "human_intervention", "target": "P5", "reason": "ambiguous_intent" }
Memory Update	{ "type": "memory_update", "scope": "private", "agent": "Research" }

07 · COMMUNICATION ARCHITECTURE

How agents communicate — with each other and with humans

Agents are completely decoupled from transport mechanisms. Rather than hardcoding `send_email()` or `dashboard_alert()`, every agent expresses a communication request abstractly. The Channel Abstraction Layer selects the appropriate delivery mechanism based on the target, urgency, and available channels. This allows new channels to be added without changing any agent logic.

Communication Abstraction Flow



Channel	Status	Use Case
Email	Active	External client communication, research delivery, alerts
Desktop Interaction	Active	Internal team notifications, simple operational decisions
Dashboard Notifications	Future	Structured operational visibility for all P-roles
Slack / Discord	Future	Team collaboration and rapid internal coordination
SMS / Tickets	Future	Escalation and incident management channels

08 · HUMAN LAYER & ESCALATION MAP

P1–P5 roles, domain ownership, and escalation routing

Humans are first-class architectural actors in this system — not emergency fallbacks. Every domain has a designated human owner, and escalation paths are deterministic: the Supervisor always routes to the correct person based on the nature of the issue.

P1	P2	P3	P4	P5
• OpenClaw behaviour & memory strategy	• n8n workflow design and execution	• Docker / VPS deployment	• Dashboard design and analytics views	• Client communication oversight
• Agent design and prompt architecture	• Integration and API connectivity	• Security and access control	• Operational data presentation	• Research consumption and decisions
• Supervisor logic and routing rules	• Automation layer reliability	• Monitoring and observability	• User interface for team interactions	• Helpdesk and marketing oversight
• AI/ML architecture decisions	• Execution middleware management	• Health escalations and incidents	• Future visualization layer	• Business operations and strategy

Escalation Routing Examples

Issue Type	Route	Escalated To
Lead qualification ambiguity	Supervisor	P5
Service / business question	Supervisor	P5
Marketing strategy decision	Supervisor	P5
n8n workflow failure	Supervisor	P2
Infrastructure / VPS issue	Supervisor	P3
Security incident	Direct alert	P3 + Support Mail
AI logic / prompt behaviour	Supervisor	P1
Dashboard / UI issue	Supervisor	P4

09 · DATABASE & INFRASTRUCTURE

Data persistence, deployment stack, and infra decisions

MongoDB Primary (v1 Active)	Conversations, memory payloads, research reports, dynamic documents, flexible schema data. Existing implementation continues. Chosen for schema flexibility appropriate to the evolving nature of v1 data models.
PostgreSQL Future Addition	Structured operational data, analytics, observability, workflow state persistence. Added when operational maturity justifies relational constraints and richer query capability.

Infrastructure Stack

Docker	Container orchestration for all services
VPS	Cloud deployment host for the entire stack
Nginx	Reverse proxy, SSL termination, routing
Monitoring	Service health visibility and alerting
Observability	Distributed tracing and log aggregation
Auth Handling	Managed by n8n / infra layer, never by agents

10 · FRAMEWORK STRATEGY & DEPLOYMENT DIRECTION

How this system generalizes to client deployments

Reusable Framework, Custom Architectures

The decision is explicitly against a universal template system. Instead, the Syntrixa system establishes a reusable orchestration framework — covering memory models, communication abstraction, execution contracts, infra patterns, and governance — which is then assembled into a custom agent graph for each deployment.

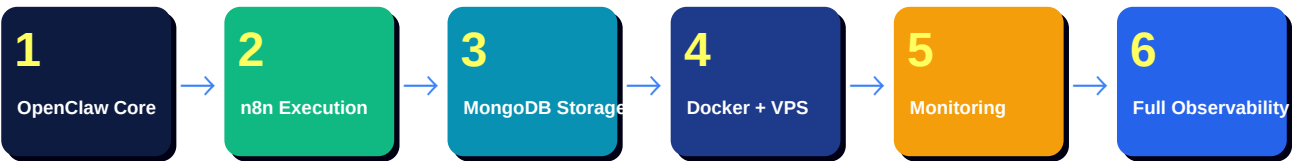
Reusable Framework Components	Customized Per Deployment
Orchestration & routing model Communication abstraction Memory systems Execution layer (n8n) Governance & guardrails Infra patterns	Agent graph design Agent specializations Domain ownership map Tool ownership per agent Escalation routing Knowledge bases

Example Future Deployments

Manufacturing Client	Production · Inventory · Maintenance · Operations · QA
HR / Recruitment	Recruitment · Onboarding · Compliance · Analytics · Support
SaaS Company	Customer Success · Sales Ops · Technical Support · Growth · Dev Ops
E-commerce	Order Management · Customer Service · Marketing · Logistics · Finance

Incremental Deployment Stack

Build Order — Near Production Stack



The system is built incrementally toward the full vision. Each layer is production-quality when shipped — not a placeholder. The goal is to learn through actual deployment, acquiring real infrastructure and orchestration experience at every step.